Demand Forecasting & Inventory Optimization System

Demand Forecasting & Inventory Control Using XGBoost, EOQ, ROP & Safety Stock

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Why Does Inventory Management Fail?

High SKU Complexity

1,163 unique SKUs with highly volatile daily demand patterns and high Coefficient of Variation across products

Inadequate Forecasting

Traditional methods like Moving Average and Naïve Forecast produce high errors when dealing with volatile demand

No Adaptive System

Manual restocking decisions lead to overstock or understock situations without per-SKU reorder point calculation

Stockout Risk

Absence of safety stock mechanism creates risk of stockout during demand spikes or supply delays

Dataset Overview: 296,110 transaction records spanning September 2022 through September 2024

Data-Driven Questions We Answered

1

Demand Volatility Analysis

Question: How volatile is daily and monthly demand per product?

Finding: Coefficient of Variation analysis reveals high demand uncertainty requiring robust forecasting methods

2

Pattern Recognition

Question: Are there seasonal or trend patterns in demand?

Finding: Month and day-of-week patterns confirmed, justifying temporal feature engineering for ML models

3

Product Diversity

Question: How diverse are product characteristics across inventory?

Finding: 1,163 SKUs across multiple categories and suppliers require individualized forecasting approaches

4

Risk Assessment

Question: Is stockout risk real without safety stock mechanism?

Finding: Historical demand analysis confirms significant stockout risk, justifying ROP and safety stock design

5

Feature Validation

Question: Can lag and rolling features predict demand accurately?

Finding: Lag-1 and Lag-2 autocorrelation confirmed, justifying XGBoost feature engineering strategy

Dataset: rekap_penjualan.xlsx

296K

Transaction Records

Complete sales data captured over 24 months

1,163

Unique SKUs

Product codes spanning multiple categories

24

Month Period

September 2022 through September 2024

Dataset Structure

Column Name	Type	Description
Tanggal	Date	Transaction date
KODE	Integer	Product code
NAMA BARANG	String	Product name
SATUAN	String	Unit of measure
KATEGORI	String	Product category
SUPPLIER	String	Supplier name
QTY	Integer	Quantity sold

System Architecture: From Data to Decision

01

Historical Sales Data

`rekap_penjualan.xlsx`

02

EDA & Feature Engineering

Lag features, rolling stats, temporal features

03

XGBoost Demand Forecasting

Per-SKU daily prediction

04

EOQ / ROP / Safety Stock

Inventory optimization calculation

05

Streamlit DSS Dashboard

Interactive simulation and decision support

This end-to-end architecture transforms raw transaction data into actionable inventory decisions through systematic data processing, machine learning forecasting, and inventory optimization calculations.

XGBoost for Demand Forecasting

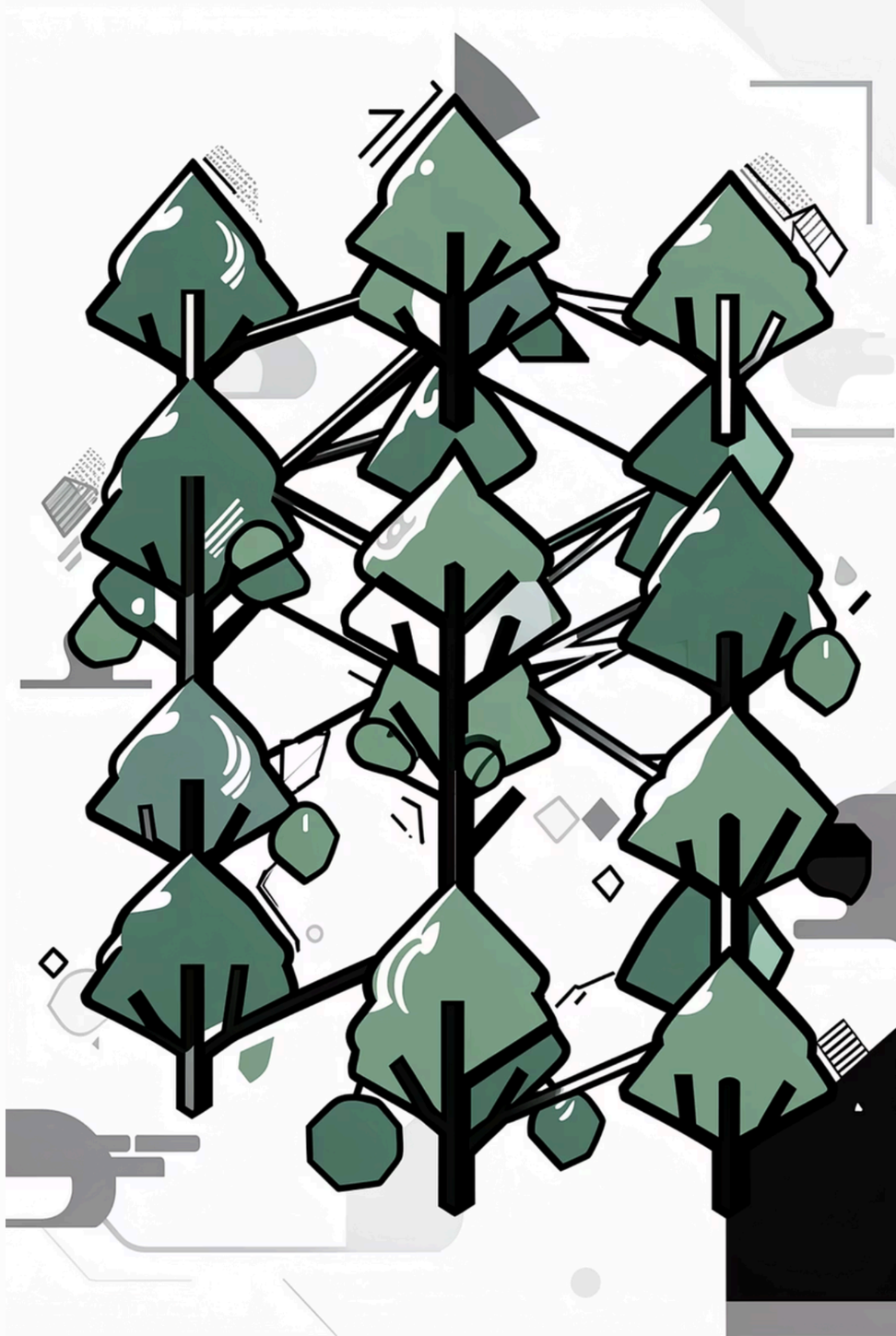
Key Features Used

- Lag_1_days, Lag_2_days, Lag_7_days for temporal patterns
- Rolling_Mean_3, Rolling_Mean_7 for trend detection
- Rolling_Std_3, Rolling_Std_7 for volatility measurement
- Month, DayOfWeek as temporal features
- Encoded KODE, KATEGORI, SUPPLIER for product characteristics

Why XGBoost?

- Handles non-linear demand patterns effectively
- Robust to outliers in FMCG data
- Captures temporal autocorrelation through lag features
- Outperforms Naïve and Moving Average baselines
- Scalable to 1,163 SKUs simultaneously

Performance: XGBoost achieved lower MAPE and RMSE compared to Naïve, MA-7, and MA-30 baseline methods



The Math Behind the System

Safety Stock (SS)

$$SS = Z \times \sigma_d \times \sqrt{L}$$

Z = Z-score (1.64 for 95% service level)

σ_d = Standard deviation of daily demand

L = Lead time in days

Buffer inventory to prevent stockouts during demand variability

Reorder Point (ROP)

$$ROP = (\mu_d \times L) + SS$$

μ_d = Average daily demand

L = Lead time (days)

SS = Safety Stock

Triggers restocking before stockout occurs

Economic Order Quantity (EOQ)

$$EOQ = \sqrt{\frac{2DS}{H}}$$

D = Monthly demand

S = Ordering cost

H = Holding cost per unit

Optimal order quantity minimizing total inventory costs

Service Level: 95% | **Z-Score:** 1.645 applied uniformly across all 1,163 SKUs

Live Simulation: AQUA BTL 600ML (SKU 695)

Input Parameters

5
Lead Time
Days to receive order

80
Initial Stock
Starting inventory units

31
Simulation Period
Days in May 2026

Simulation Results

462
Total Forecast Demand
Expected sales over period

15.4
Avg Daily Demand
Units per day

2
Orders Placed
Restocking events

390
EOQ Quantity
Units per order

Timeline

Day 1
Stock: 57 units remaining → Reorder Required → Order Placed

1

2

Days 2-5
Awaiting Order Arrival (lead time processing)

3

Day 6+
Order arrived → Stock replenished → Sufficient inventory

Streamlit DSS: What Users Can Do

Overview Page Features

- 📊 Sales trend analysis (daily and monthly views)
- 📈 Demand distribution histograms for distribution insights
- 🏆 Top 10 best-selling products with bar chart visualization
- 🍕 Sales by category breakdown with pie charts
- 🔍 Dynamic filtering by date range and product category

Inventory Simulation Page

- ✅ Select any of 1,163 SKUs for analysis
- ⚡ Auto-calculated EOQ from historical demand patterns
- ⚙️ Configurable lead time and initial stock parameters
- 🔄 ROP simulation with daily inventory tracking
- 🚦 Status flags: Reorder Required / Awaiting / Sufficient
- 💾 Download simulation results as CSV for record-keeping



Key Takeaways & Next Steps

What We Achieved

- ✅ Built data-driven inventory DSS for 1,163 SKUs
- ✅ XGBoost outperforms traditional forecasting baselines
- ✅ EOQ + ROP + Safety Stock reduces stockout risk at 95% service level
- ✅ Interactive Streamlit dashboard enables real-time decision support
- ✅ System deployable and extensible for retail/FMCG applications

Future Improvements

- 🔧 Integrate real-time POS data feed for live updates
- 🔄 Add LSTM/Transformer models for long-horizon forecasting
- 🌐 Multi-echelon inventory optimization across locations
- 📊 Dynamic service level per category (ABC classification)
- 💰 Cost optimization dashboard (holding + ordering cost minimization)

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Open Dashboard